Fight Fire with AI

Project Workbook

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Chapter 1. Literature Search, State of the Art

Literature Search

Climate change has led to many unexpected and disastrous consequences, particularly the likelihood of wildfires being increased. In fact, Ye et al. (2026) reports in their study, “Wildfires and Public Health: A Comprehensive Review of Human-Centric Studies” (2026), published under GeoHealth, that in the past decade, wildfires in particular have dealt massive damage, with the United States in particular having experienced over 530,000 wildfire incidents in the past decade, burning more than 70 million acres of land. Even more concerning is the effect wildfires and generally poor air quality can have on human physiology, oftentimes not just leading to cardiovascular and respiratory issues, but even psychiatric conditions like anxiety, PTSD, and depression. While the dangers of wildfire are concerning, many organizations have done much to address these dangers, developing frameworks and AI models for risk assessment, early warning systems, and public health interventions aimed at reducing wildfire exposure and its downstream health consequences.

In the study, “Using AutoML and generative AI to predict the type of wildfire propagation in Canadian conifer forests” published under Ecological Informatics, Khanmohammadi et al. (2024) looks at Canadian fires and aim to see what the best ML method is for studying wildfire prediction and prevention. They discover that TabPFN outperforms other ML methods in this context, naming it as having the most potential to support fire management, specifically in the context of fuel management and identifying high fire spread along with intensity potential.

In terms of simulation, “FORFIS: A forest fire firefighting simulation tool for education and research”, Bredlau et al. (2023) presents a Python based simulation software that models wildfires using squares and hexagons. The model uses math equations and probability functions to map the spread of wildfires to the surrounding area, giving it potential to predict wildfire spread sooner to help with evacuation planning in case of emergency. Additionally, Linn et al. (2020) in their study, “QUIC-fire: A fast-running simulation tool for prescribed fire planning”achieves a

similar tool by using authors use numerical methods to compute solutions to differential equations which model how fires move, specifically the Navier-Stokes equations. Although wildfires can be seemingly sporadic, mathematical models can be computed to make predictions that would benefit cases of emergency.

Although moments of emergency are important to address, wildfires also lead to lingering and devastating effects that can be addressed with the help of data analysis. In particular, “Big data analyses for determining the spatio-temporal trends of air pollution due to wildfires in California using Google Earth Engine”, Al Saim and Aly (2024) use data analysis to identify historical spatiotemporal trends of air pollution attributable to wildfires in California. With this study and similar studies in mind, this approach also gives potential to prevention and mitigation programs, as well as informing residents with preexisting health conditions.

Simulation Software:

https://digitalcombustion.com/, FlamMap, https://wildfire.concord.org/

State of the Art Summary

Based on current academic research, the state-of-the-art in wildfire prediction and monitoring have moved beyond simple visual detection, focusing instead on advanced predictive modeling, edge computing, and multi-modal data fusion techniques. Some of the state-of-the-art tools in wildfire prediction and monitoring include:

- **Deep Time-Series Networks for Orbital Data:** conventional spatial image processing is no longer used in favor of temporal deep learning techniques. Deep Gated Recurrent Unit (GRU) networks are now used for processing massive time-series data from geostationary satellites such as GOES-R. Such techniques enable systems to detect pixel-level emergence of thermal anomalies much earlier than conventional single image-based wildfire detection techniques (Zhao et al., 2021).

- **AutoML and Generative AI (GANs)**: In terms of predictive modeling, a critical issue is that of a scarcity of empirical wildfire data. However, the current best practice is to employ Generative Adversarial Networks (GANs) to generate artificial training data for infrequent extreme wildfire events. Subsequently, these generated datasets are then fed into AutoML tools like TabPFN for hyperparameter optimization in order to make accurate predictions regarding wildfire propagation behavior (Khanmohammadi et al., 2024).
- **Edge Computing and Resource-Constrained AI:** Rather than relying on computationally intensive and high-latency cloud computing for processing, current best practices in wildfire modeling employ resource-constrained deep learning models for direct deployment onto edge computing platforms such as Unmanned Aerial Vehicle (UAVs) and Unmanned Ground Vehicles (UGVs). Powered by microcomputer architectures such as Raspberry Pi, these models employ multi-sensor fusion techniques to enable autonomous path planning and fire segmentation in the field (Chen, 2025; Das et al., 2026).
- **Unified Detection Emission Pipelines**: The trend in the industry is moving towards considering fire detection and post-fire analysis as one continuous process or pipeline. Modern computer vision systems are specifically designed to output precise spatial parameters(for instance, accurate burned area marks) to be used to drive mathematical emission quantification models with high degrees of precision (Das et al., 2026).

References

1. Ye, X., Ye, Y., Huang, X., & Onega, T. (2026). Wildfires and Public Health: A Comprehensive Review of Human‐Centric Studies. *GeoHealth*, *10*(2). https://doi.org/10.1029/2025gh001534

   This systematic review examines the impacts of wildfire on human health across physical, psychological and social dimensions. Ye et al. identifies uneven research coverage across different populations and study designs, mostly looking into mental health and demographic subgroups. They highlight the need for innovative machine-learning approaches to advance wildfire-health research.

2. Sadegh Khanmohammadi, Cruz, M. G., Daniel D.B. Perrakis, Alexander, M. E., & Mehrdad Arashpour. (2024). Using AutoML and generative AI to predict the type of wildfire propagation in Canadian conifer forests. *Ecological Informatics*, *82*, 102711–102711. https://doi.org/10.1016/j.ecoinf.2024.102711

   This study develops and evaluates machine learning models for predicting wildfire propagation time, being able to distinguish between surface fires, passive crown fires, and active grown fires, all with the help of experimental figure data from Canadian forests.

3. Bredlau, M., Weber, A., & Knoll, A. (2023). FORFIS: A forest fire firefighting simulation tool for education and research. Retrieved March 5, 2026, from arXiv.org website: https://arxiv.org/abs/2305.17967

   Bredlau et al. presents an open-source forest fire firefighting simulation tool implemented with the help of Python. The tool features a user-friendly interface and a simulation engine that can be modified based on cellular automata,

available in both rectangular and hexagonal cell decompositions to model fire spread.

4. Linn, R. R., Goodrick, S. L., Brambilla, S., Brown, M. J., Middleton, R. S., O'Brien, J. J., & Hiers, J. K. (2020). QUIC-fire: A fast-running simulation tool for prescribed fire planning. *Environmental Modelling & Software*, *125*, 104616. https://doi.org/10.1016/j.envsoft.2019.104616

   Linn et al. introduces QUIC-Fire, a simulation tool that models itself after fire atmospheres and is prescribed for fire planning. QUIC-Fire accepts 3D fuel structure inputs and models local wind-fire data, making it practical for ensemble-based prescribed burn planning without the need for high-performance computing resources.

5. Saim, A. A., & Aly, M. H. (2024). Big data analyses for determining the spatio-temporal trends of air pollution due to wildfires in California using Google Earth Engine. *Atmospheric Pollution Research*, *15*(9), 102226. https://doi.org/10.1016/j.apr.2024.102226

   The study uses Google Earth Engine with MODIS, Sentinel-5P, and AERONET satellite datasets to evaluate how California wildfires impact air quality. Key findings include a maximum correlation between wildfire activity and air pollution indicators, increased nitrogen dioxide, and a rising trend in Aerosol Optical Depth attributable to wildfires.

6. Chen, H. (2025) Development of an autonomous unmanned ground vehicle system for forest fire detection and exploration with multi-sensor fusion and web-based control. 2025 *International Conference on Intelligent Control and*

Electrical Engineering (IC-ICEE), 1-6. Doi: 10.1109/IC-ICEE66522.2025.11200222.

This conference paper discusses the architecture of an autonomous UGV that was field-tested for safe exploration of fire scenes. The UGV was developed using the Raspberry Pi microcomputer, which enables the use of multi-sensor fusion technology that combines ultrasonic and infrared sensors for obstacle avoidance with computer vision technology for fire and smoke recognition. The author of this paper highlights the high degree of success that this UGV achieves in autonomous navigation, including the use of low-latency web-based interface for safe remote viewing of telemetry data from the server.

7. Das, K. Poovvancheri, J., Flesca, S. Calidonna, C.R., & Chen, D. (2026) Emerging trends in wildfire detection through the lens of computer vision and wildfire emission quantification: A comprehensive survey. *IEEE Access,* 14, 20205-20228. doi: 10.1109/ACCESS.2026.3660843.

   In this detailed review, the authors discuss the important intersection between modern computer vision-based detection systems and subsequent environmental emission models. The paper highlights the industry move toward the use of lightweight deep learning architectures on resources-constrained devices, such as UAVS, in order to minimize cloud transmission latency. Significantly, the authors note that in order for the output of a detection model, such as burned area masks and Fire Radiative Power, to be useful, it must be easily integrated into emission model pimples in order to minimize spatial uncertainty in climate and air quality assessments.

8. Zhao, Y., Ban, Y., & Nascetti, A. (2021). Early detection of wildfires with GOES-R time-series and deep GRU network. *IGARSS 2021 - 2021 IEEE International Geoscience and Remote Sensing Symposium*, 97-100. doi: 10.1109/IGARSS47720.2021.9554225.

   The current research aims to address the spatial-temporal trade-off in wildfire monitoring using satellites through the exploitation of the high temporal resolution of 15 minutes between consecutive images from GOES-R satellites in geostationary orbit. The research proposes a time series-based Deep gated Recurrent Unit (GRU) network that can process up to 20 different timestamps simultaneously for active wildfire segmentation in terms of pixels. The researchers conclude that their temporal deep learning method can locate emerging wildfires two hours before existing single-image-based products, such as those from VIIRS(Visible Infrared Imaging Radiometer Suite) from other satellite sensors, thereby reducing false alarms and missed alarms.

Chapter 2. Project Justification

This project is justified by the level of wildfires seen in the past few decades, as well as the effects wildfires have on the environment outside of what can be clearly seen. Wildfires can be hard to detect until it's too late, at which point it takes special care and outside forces to bring an end to the damage that's caused. They can be costly to put out and detect without proper treatment and attention. We believe we can work towards finding a solution to the problem with the help of AI developments, as well as some hardware and software solutions by the use of the Raspberry Pi.

AI has made its impact into many fields and cutting edge technologies over the past few years. AI agents are developed for the sake of making things easy for people and allowing them to do things that weren't possible as recently as 10 years ago. Through all the recent developments of AI, we believe that it will be an essential tool that we will use to address the issue of fighting wildfires, and reconciling the extraneous environmental impacts that result.

We believe this project will be unique in how it tackles an ongoing issue using new methodologies and technological developments. We understand that similar ideas have been presented in other projects that aim to fight wildfires, but we wish to go beyond by understanding things from a broader perspective and attack the problem on multiple levels. By looking at environmental factors that result from wildfires we hope to make an impact on not only wildfire prediction and mitigation, but also other natural disasters that may result from raging wildfires.

Chapter 3. Project Requirements

Requirements

User Stories:

- As a user, I want to visualize the spread of a wildfire within a region in order to see areas highly susceptible to wildfires.
- As a user, I want the best road route to avoid an incoming fire.
- As a user, I want to be able to view a dashboard of today and this week's Fire prediction
- As a user, I want to see how many fire responders are on the scene currently so that I can understand whether to call the police or contact my family.
- As a user, I want to see the probability and confidence for the predicted danger zone so I can trust how much to trust the fire forecast
- As a user, I want to understand what particle matter is in the wildfire as I may have health conditions that make my respiratory system sensitive or vulnerable.
- As a user, I want to be able to see a prediction of any wildfires for the upcoming week.
- As a user, I want to input my health profile (elderly, pets, asthma), so that I can receive a personalized wildfire and air-quality risk score or label.
- As a user, I want a wildfire vulnerability score for where I am located based on vegetation density, historical fires, and
- As a user, I want to be able to fine tune my notifications and alerts.
- As a user, I want to be able to see nearby evacuation shelters, safe zones, and medical centers near me in case of an emergency.
- As a user, I want to be able to report visible smoke, flame, or road blockages to improve real-time situational awareness.
- As a user, I want to simulate how projected temperature increases or drought conditions would affect wildfire probability in my region.

Functional Requirements(FR)

I. Essential Features

A. **FR-E01 [Prediction Visualization]:** Visualize current fire perimeters (derived from 10-minute GOES-18 data) and deep learning-predicted spread polygons on the interactive map.

B. **FR-E02 [Dynamic Routing]:** Calculate optimal evacuation routes avoiding roads that intersect with predicted fire danger polygons.

C. **FR-E03 [Confidence Metrics]:** Display the deep learning model's probability percentage and confidence intervals for forecasted danger zones.

D. **FR-E04 [Telemetry & Environmental Hazards]:** Display live telemetry (AQI, Wind Speed/Direction), compute vulnerability scoring, render Deck.gl animated wind particle arrays reflecting live vectors, and overlay NWS Red Flag Warning polygons when active.

E. **FR-E05 [Emergency POIs]:** Display static Points of Interest using custom HTML markers (e.g., FEMA Evacuation Shelters with Lucide-react icons) anchored securely to the map. Utilizes downloaded, open-source FEMA/CalOES datasets loaded into PostGIS.

F. **FR-E06 [Alert Configuration]:** Allow configuration of SMS/push notifications for approaching thermal anomalies or risk polygons.

G. **FR-E07 [Context-Aware Chat]:** Provide a simulated RAG-backed chatbot interface answering queries with contextual health data (e.g., asthma risks, AQI, shelter routing) and explicit citations (e.g., NWS, CalOES, BAAQMD).

H. **FR-E08 [Source Citation]:** The chatbot must explicitly cite the source document (e.g., CalFire, PubMed) retrieved from the vector database.

II. Desired Features

A. **FR-D01 [Extended Forecast]:** Generate a 7-day predictive fire risk map based on forecasted weather covariates.

B. **FR-D02 [Air Quality Integration]:** Overlay real-time PM2.5 and PM10 IoT data on the map interface.
C. **FR-D03 [Health-Personalized Risk]:** Generate personalized risk labels (e.g., "High Risk for Asthma") based on user-inputted health profiles and local AQI.
D. **FR-D04 [Conversation History]:** Maintain chat session context to allow for seamless follow-up questions.

III. Optional Features

A. **FR-O01 [Climate Simulation]:** Provide a sandbox interface to tweak temperature/humidity parameters and re-run the ConvLSTM model to visualize simulated probabilities.
B. **FR-O02 [Crowdsourced Reporting]:** Allow users to submit geolocated pins for visible smoke or road blockages.
C. **FR-O03 [First Responder Tracking]:** Map general vicinity of dispatched fire units (dependent on open-source CAD data availability).

Non-Functional Requirements(NFR)

I. Performance & Latency

A. **NFR-P01 [Inference Speed]:** The Python deep learning service shall return a predicted fire spread polygon within **5.0 seconds** of receiving the preprocessed GOES-18 and weather tensors from Django.
B. **NFR-P02 [Routing Speed]:** The GeoDjango routing service shall recalculate an A* path avoiding dynamic fire polygons across the Bay Area road graph in under **2.0 seconds** for distances up to 100 miles.
C. **NFR-P03 [Streaming Latency]:** The Django Channels chat interface must stream the RAG response with a "Time to First Token" (TTFT) not exceeding **2.0 seconds**.

II. Scalability

A. **NFR-S01 [Concurrent Users]:** Django Channels and the Redis broker shall support a minimum of **5,000 concurrent user connections** without exceeding a 95th percentile response time of 500ms.

B. **NFR-S02 [Safety & Guardrails]:** The RAG pipeline must implement a strict system prompt constraint preventing it from diagnosing medical conditions.

III. Reliability & Fault Tolerance

A. **NFR-R01 [Graceful Degradation]:** If external weather APIs fail, the pipeline shall automatically fall back to utilizing historical seasonal averages for the Bay Area.

B. **NFR-R02 [LLM Fallback]:** If the external LLM API times out, the system shall disable the chat widget and display a "Service Unavailable" badge without disrupting core routing.

C. **NFR-R03 [Availability]:** The core API and mapping services shall maintain **99.9% uptime** during simulated disaster stress-tests.

IV. Security & Privacy

A. **NFR-SEC01 [Data Protection]:** Health profile data shall be anonymized, encrypted at rest using AES-256 in the database, and never transmitted to third-party APIs.

B. **NFR-SEC02 [Authentication]:** User accounts shall be secured using standard OAuth 2.0 or JWT (JSON Web Token) authentication protocols.

Chapter 4. Dependencies and Deliverables

Dependencies

- *Data Synchronization* - Reliable API pipelines to integrate and align disjoint datasets such as air quality index records or geo spatial imagery
- *Data Frequency* - Check for data collection frequency and temporal resolution so that the intervals can be updated to match accurate, real-time environmental data
- *Data Quality* - Getting good satellite imagery that can be understood using CNNs and show the spread of wildfires over time through physical features
- *Mathematical Methods* - Defining and understanding partial differential equations (PDE) along with heat equations to predict fire spread in real time

Deliverables

<u>Mathematical Wildfire Prediction Modelling (MWPM)</u>

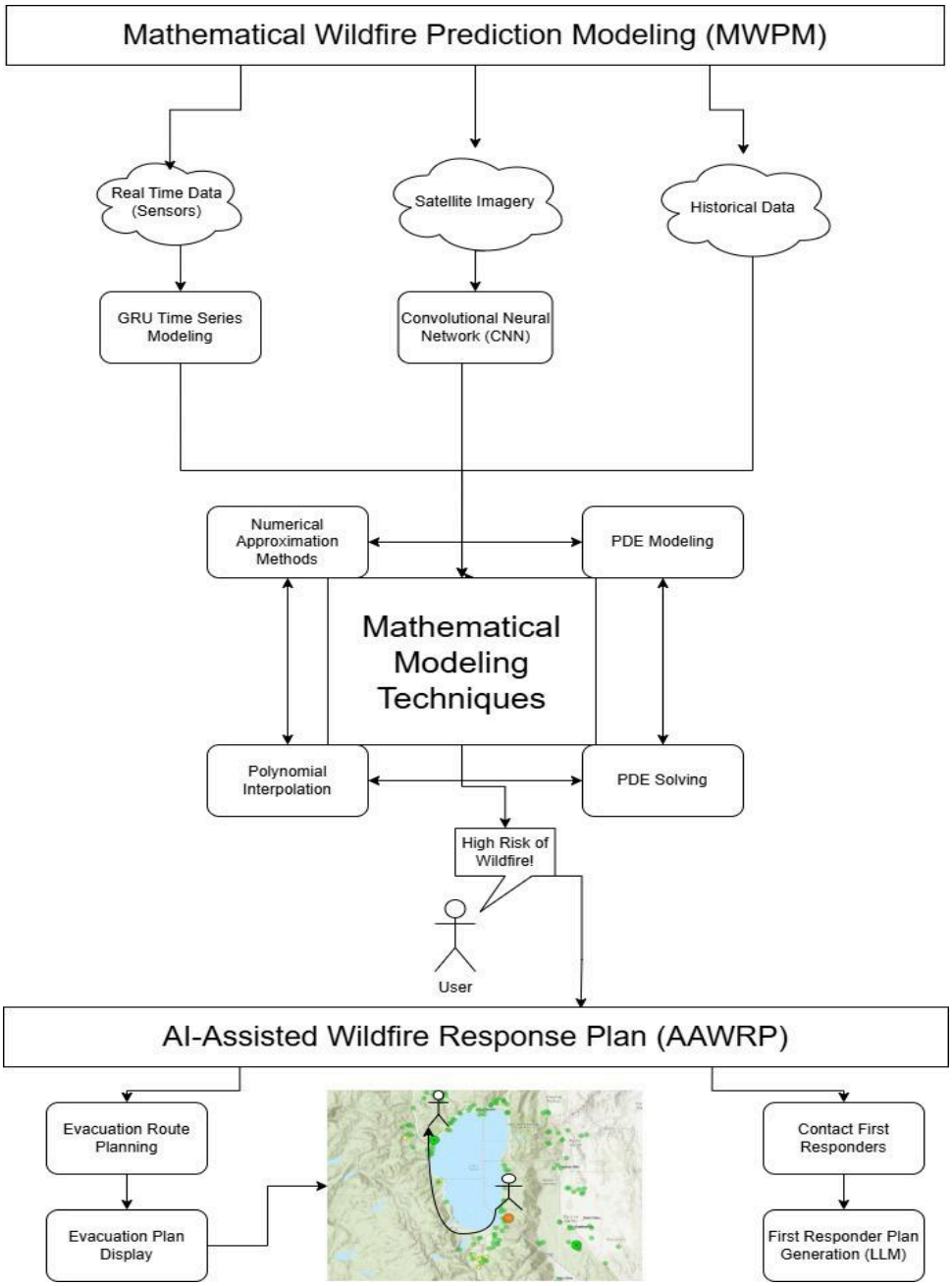
- *Dashboard* that shows a map of nearby places of high wildfire risk
- *Web Application* that gives updates and possible evacuation plans from areas of high risk to low risk
- *Mathematical Modeling Methods and PDE Solver* for the purpose of mathematically predicting places of high/low risk

<u>AI-Assisted Wildfire Response Plan (AAWRP)</u>

- *AI-Assisted Routes* from places of high risk to low risk, using path-finding methods
- *AI-Assisted Methods for Fighting Wildfires*, giving first responders specific instructions on how to navigate and prevent the wildfire from causing any more damage

Chapter 5. Project Architecture

Version 1.



Architecture Description:

As can be seen in the diagram, there are two main stages of the architecture: *Mathematical Wildfire Prediction Modeling (MWPM)* and *AI-Assisted Wildfire Response Plan (AAWRP)*.

Mathematical Wildfire Prediction Modeling (MWPM)

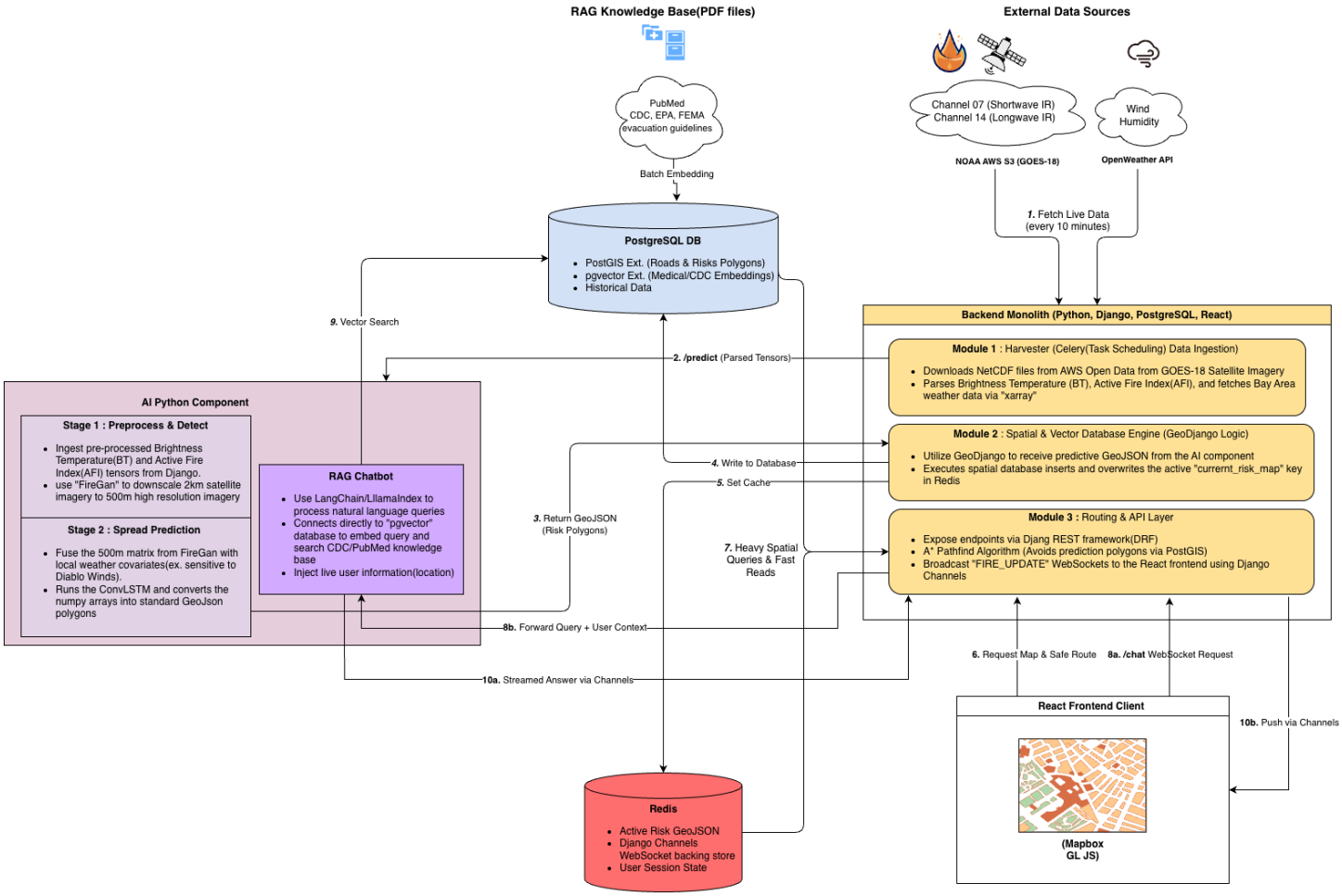
In this stage, we focus on the collection of data through various sources, and aim to use the data for the purpose of predicting and modeling wildfire behavior as best as we can. We look at historical wildfire data, satellite imagery, and real time data of important features captured by sensors or other instruments. Considering this data comes from multiple sources, we aim to synchronize everything to be useful for us. We can use **Convolutional Neural Networks (CNNs)** to extract information from satellite imagery and **GRU time-series modeling** to help us analyse our real time data from sensor-based tools. With historical data there can be a lot to look over which may be helpful for us, but we aim to extract the most important features in an efficient way.

Once we have all the data available, we use various mathematical tools to unpack the data and create a mathematical model which can predict the risk of wildfires occurring. We can predict wildfire behavior using each important feature, by seeing how each feature affects each other in the context of wildfires starting. There are many methods of creating mathematical models, the two that we most strongly consider are **polynomial interpolation** and **partial differential equations**, most specifically the heat equation. The heat equation tells us that the rate of change of the temperature at a given point in space is proportional to the laplacian (space curvature) of the temperature at that same point in space. We can take the idea of space and time derivatives of temperature affecting each other, and use numerical methods to both create and solve PDEs that model our temperature flow. Polynomial interpolation is another method which can tell us the temperature (or another such feature) at a point in between two points we already know. Both of these methods can work in tandem with each other, and ideally will give us models that allow us to take all our data we collect, and predict various unknown quantities. If a wildfire is predicted to happen, we move onto the next stage of the architecture, *AAWRP*.

AI-Assisted Wildfire Response Plan (AAWRP)

For this stage, we employ methods to mitigate the damages caused by wildfires. We both aim to get affected users to safety by generating and displaying an evacuation plan, as well as creating a plan to fight the wildfire which we give to first responders. The evacuation plan we can create using path finding methods to find the most efficient route from a place of high risk to a place of low risk, then display the route on the interactive dashboard for the user. For fighting the wildfire, we can generate a plan against the wildfire using an **LLM**, and use various tools to find and get in contact with first responders and trained firefighters, then create a clear and comprehensive strategy with specific instructions for them to follow.

Version 2.



Component Specification:

- ***Django Monolith (Web & Orchestration)***
  - **Module 1 (Data Harvester):** Executes a 10-minute task via Celery. Queries AWS Open Data, downloads the actual NetCDF file, and uses “xarray” to spatially crop the data to the Bay Area bounding box. Simultaneously fetches live wind data (OpenWeather), PM2.5 Smoke Conditions (OpenAQ) and active Red Flag Warnings (NWS API). It pre-calculates the Brightness Temperature (BT) and Active Fire Index (AFI) tensors, then POSTs these preprocessed arrays to the AI Engine Component.
  - **Module 2 (Database Engine):** Utilizes GeoDjango to receive predictive GeoJSON from the AI. Executes spatial database inserts for historical tracking and overwrites the active “current_risk_map” key in Redis. It will also be

responsible for serving static FEMA/CalOES designated shelter points from POSTGIS.
    - **Module 3 (API/Routing):** Exposes endpoints via Django REST Framework (DRF). Uses user GPS coordinates to query PostGIS via GeoDjango for A* pathfinding across the Bay Area road graph. Broadcasts "FIRE_UPDATE" WebSockets to the React frontend using Django Channels.
- ***Python Inference Engine (FastAPI)***
    - **Stage 1 (Detection):** Receives the pre-cropped 2km Bay Area matrices directly from Django. Instantly runs the FireGAN model to downscale and output 500m resolution arrays.
    - **Stage 2 (Prediction):** Fuses 500m matrix with local weather covariates (sensitive to local phenomena like Diablo Winds). Runs the ConvLSTM and returns standard GeoJSON polygons.
    - **Rag Chatbot:** Exposed the LLM interface. Takes the natural language query, embeds it, queries PostgreSQL (pgvector), and streams the contextually aware response (enriched with live PM2.5 and Red Flag status) back through the Django backend.

Tech Stack Breakdown:

- **FrontEnd (UI/UX):** React.js Mapbox GL JS (via react-map-gl for rendering complex GeoJSON polygons), Deck.gl (WebGL overlay for dynamic animated wind particles).
- **Backend (Core Monolith):** Python, Django REST Framework, Celery (Task Scheduling).
- **Database:** PostgreSQL
    - Extension 1: POSTGIS (Managed via GeoDjango. Contains only the Northern California/Bay Area ".osm.pbf" extract to ensure A* pathfinding executes under 2.0 seconds).
    - Extension 2: pgvector (For native vector storage and semantic similarity searches).
- **Caching & Live State:** Redis (Serves triple-duty as the Celery task broker, the Django Channels WebSocket backing store, and the high-speed GeoJSON cache).
- **(Flexible) AI/Inference Engine:** Python, FastAPI, PyTorch (FireGAN/ConvLSTM trained on California fuel types like Chaparral), LangChain/LlamaIndex

External Data Sources:

To achieve low-latency detection and prediction within the Bay Area, the system relies on:

- **Primary Live Feed Satellite Imagery (NOAA GOES-18):** Advanced Baseline Imager (ABI) data accessed via AWS Open Data Registry. Downloaded and preprocessed by Django (ingesting Channel 07 and Channel 014) every 10 minutes.

- **Secondary Feed (Weather Covariates & AQI):** OpenWeather/GRIDMET APIs providing localized wind speed, wind direction, and humidity. OpenAQ API or PurpleAir API providing live PM2.5 data for the Bay Area (feeding the RAG chatbot for personalized health advice).
- **Alert Feeds (NWS):** US National Weather Service (NWS) JSON API. A Django Celery worker fetches the active "Red Flag Warnings" for the Bay Area hourly, to be rendered as a translucent red layer on the frontend map.
- **(Flexible)Training Ground Truth (Offline):** NASA FIRMS/VIIRS high-resolution imagery of historical California fires (e.g., Dixie, Mosquito, LNU Lightning Complex) to train the FireGan on local biomes.

RAG Knowledge Base (Vector Database Sources):

In order to ensure the chatbot provides authoritative, verifiable, and locally relevant advice, the "pgvector" database will be pre-indexed with:

- **Regional Guidelines:** Bay Area Air Quality Management District(BAAQMD) PM2.5 guidelines and CalOES evacuation terminology.
- State Safety Rules: Calfire Defensible Space Guidelines(Zone 0, 1, and 2 rules).
- **Medical Literature:** PubMed/PMC abstracts covering "wildfire smoke respiratory impact" and "asthma exacerbation"
- The prompt sent to the LLM will always inject the user's current live data (e.g. local Bay Area AQI, distance to fire, vulnerability score) to make the retrieved documents contextually relevant.

Chapter 6. Project Design

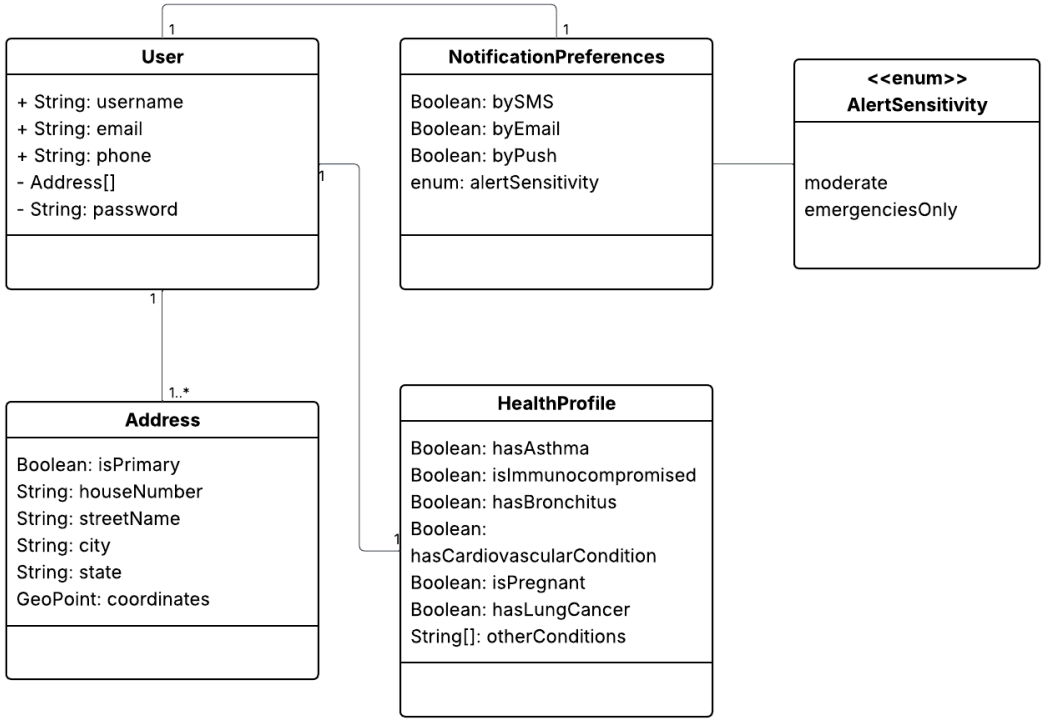
Simulation Software:

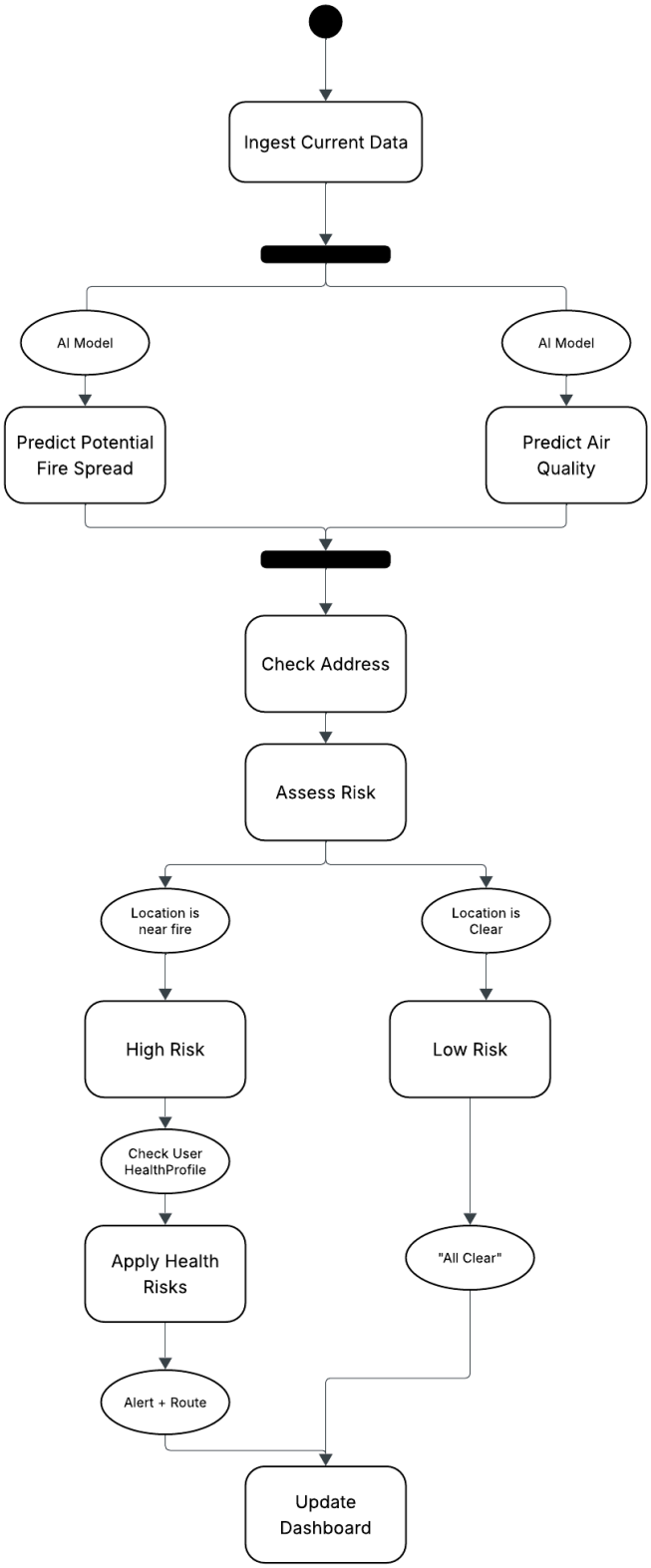
- Visualization of weather variables such as wind speed and direction
- Prediction window for wildfire spread over the next few hours
- Used to generate datasets and provide a “ground truth” for testing and validation.

AI Models:

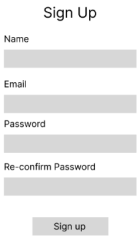
- AI that predicts wildfire risk based on real time weather data
- Unsupervised Learning that clusters data into two groups
- CNNs that analyze satellite imagery and terrain patterns
- PINN with spatiotemporal data and the Heat Equation to predict Temperature for a given position/time value
- AI/ML model sends prediction to simulation software and dashboard

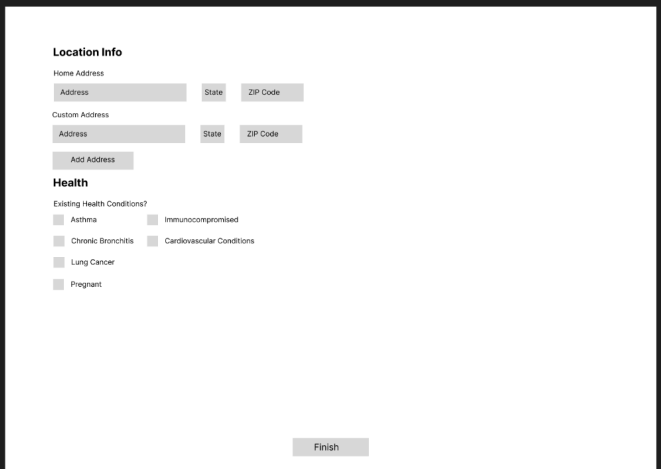
Listed below are a database diagram, a UML activity diagram, and Figma wireframe mockups.

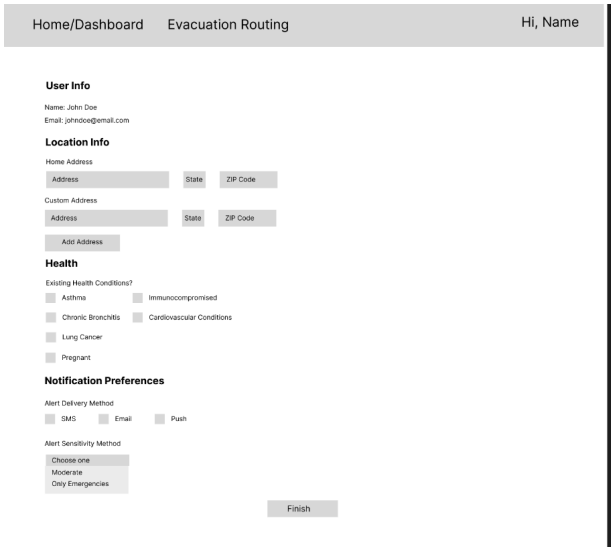

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1
User
+ String: username
+ String: email
+ String: phone
- Address[]
- String: password
NotificationPreferences
Boolean: bySMS
Boolean: byEmail
Boolean: byPush
enum: alertSensitivity
<<enum>>
AlertSensitivity
moderate
emergenciesOnly
1
1
1
1..*
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String: houseNumber
String: streetName
String: city
String: state
GeoPoint: coordinates
HealthProfile
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hasCardiovascularCondition
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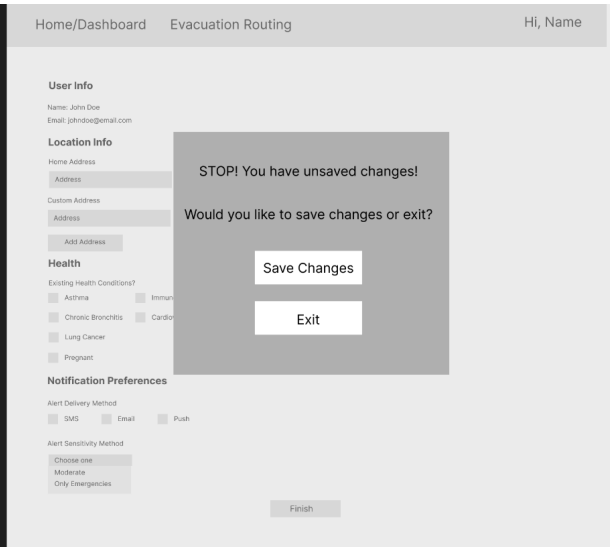

Ingest Current Data
AI Model
AI Model
Predict Potential Fire Spread
Predict Air Quality
Check Address
Assess Risk
Location is near fire
Location is Clear
High Risk
Low Risk
Check User HealthProfile
Apply Health Risks
"All Clear"
Alert + Route
Update Dashboard

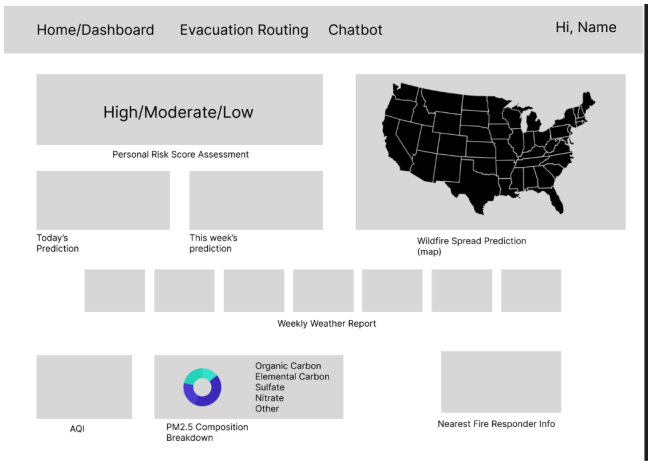

Login
Email
Password
Login
New User? Sign up

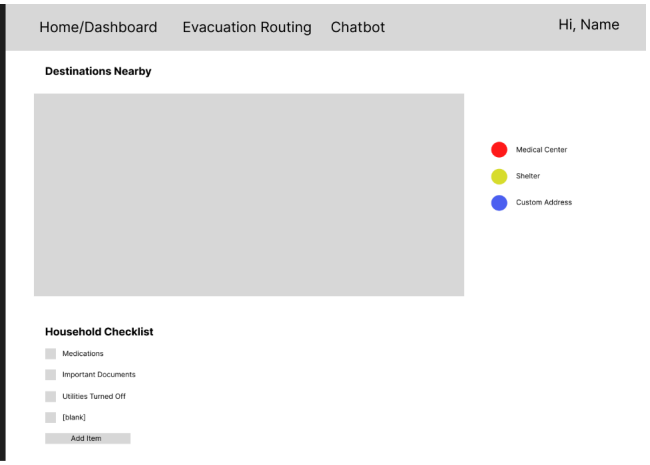

Sign Up
Name
Email
Password
Re-confirm Password
Sign up

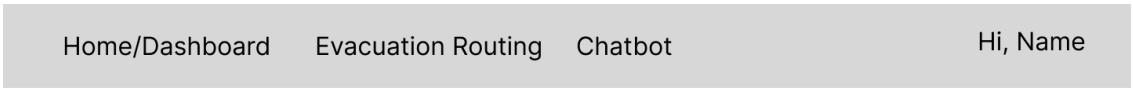

Location Info
Home Address
Address
State
ZIP Code
Custom Address
Address
State
ZIP Code
Add Address
Health
Existing Health Conditions?
Asthma
Immunocompromised
Chronic Bronchitis
Cardiovascular Conditions
Lung Cancer
Pregnant
Finish


Home/Dashboard
Evacuation Routing
Hi, Name
User Info
Name: John Doe
Email: johndoe@email.com
Location Info
Home Address
Address
State
ZIP Code
Custom Address
Address
State
ZIP Code
Add Address
Health
Existing Health Conditions?
Asthma
Immunocompromised
Chronic Bronchitis
Cardiovascular Conditions
Lung Cancer
Pregnant
Notification Preferences
Alert Delivery Method
SMS
Email
Push
Alert Sensitivity Method
Choose one
Moderate
Only Emergencies
Finish


Home/Dashboard
Evacuation Routing
Hi, Name
User Info
Name: John Doe
Email: johndoe@email.com
Location Info
Home Address
Address
Custom Address
Address
Add Address
Health
Existing Health Conditions?
Asthma
Chronic Bronchitis
Lung Cancer
Pregnant
Notification Preferences
Alert Delivery Method
SMS
Email
Push
Alert Sensitivity Method
Choose one
Moderate
Only Emergencies
Finish
STOP! You have unsaved changes!
Would you like to save changes or exit?
Save Changes
Exit


Home/Dashboard
Evacuation Routing
Chatbot
Hi, Name
High/Moderate/Low
Personal Risk Score Assessment
Today's Prediction
This week's prediction
Wildfire Spread Prediction (map)
Weekly Weather Report
AQI
Organic Carbon
Elemental Carbon
Sulfate
Nitrate
Other
PM2.5 Composition Breakdown
Nearest Fire Responder Info


Home/Dashboard
Evacuation Routing
Chatbot
Hi, Name
Destinations Nearby
Medical Center
Shelter
Custom Address
Household Checklist
Medications
Important Documents
Utilities Turned Off
[blank]
Add Item


Home/Dashboard
Evacuation Routing
Chatbot
Hi, Name




Ask Firebot a question...

Chapter 7. QA, Performance, Deployment Plan

We will deploy the software in local cities to see how well it can predict wildfires

We will see how well it navigates tricky environments to go from risky areas to safer areas

We will test and validate our data on multiple sources and datasets

We will evaluate the model against the ground truth mask to see how well it performs based on various Machine Learning metrics such as ROC-AUC, Precision, Accuracy, Recall, F1-Score

We can test different models for different datasets we find online and through other sources

In the future we will implement ConvLSTM models and other Deep Learning Architecture for more advanced predictive models

We will improve the baseline models using PINNs and other specific Neural Network architectures for example

We will create a simulation software that visualizes the spread of wildfires over specific time windows

Chapter 8. Implementation Plan and Progress

MWPM:

1. Acquire datasets related to wildfire prediction (AQI Sensors, Satellite imagery, etc.)
2. Create and evaluate models based on each dataset we find, aim to create a hybrid model
3. Improve on baseline models using advanced Deep Learning methods
4. Predict each different type of variable that relates to wildfire risk and spread through time sensitive data
5. Use Mathematical Modeling methods as necessary
6. Present data on an interactive website/dashboard that displays real time data which is used to predict wildfire risk and spread
7. Deploy and Test in real time

AAWRP:

1. Figure out how to create paths from higher to lower risk areas
2. Show paths on the web application
3. Discover best methods for fighting wildfires, likely using reinforcement learning
4. Use location data to pull up any emergency services (firefighting, shelters, etc.)
5. Design a way to relate specific instructions to said first responders

Chapter 9. Project Schedule

For our project schedule, we decided to divide into 4 two-week sprints, initially focusing on data processing and synchronization. We will then tackle the deployment of a deployable prototype. We have ensured that the work assigned to each team member is unique.

Sprint / Dates	**Milestone**	**Task Owner**	**Task Description**	**Dependencies**
Sprint 1: Data & Setup	Data Synch & Env Setup	Earl	Finding data and imagery related to NASA GOES-R and GOES-T satellites.	Data Synchronization
		Isla	Refine datasets related to EPA AQS and Purple Air or Cali Air Resources.	Data Synchronization
		Nikhil	Understand the data and extract features using API pipelines such as Amazon Greengrass and AWS S3 for information handling	Data Access
		Scott	Research and define parameters for	

			mathematical modeling.	
Sprint 2: Core Modeling	Math & AI software Prototypes	Earl	Potentially design a user friendly web application	Map Data
		Isla	Build an application repository and database schema for data storage and analysis	Sprint 1 Data
		Nikhil	Train transformer models or Convolutional Neural Network (CNN) on features like humidity and AQI for spatial temporal imaging.	Sprint 1 Data
		Scott	Implement Polynomial Interpolation and PDE solver/Heat Equation model for fire spread simulation.	Math Research
Sprint 3: Integration	System Integration	Earl	Integrate frontend web dashboard with fire spread and air quality	Sprint 1 & 2

			visualization in the backend ML APIs.	
		Isla	Backend API development to connect the Dashboard to the Math/AI models	Sprint 2
		Nikhil	Optimize the CNN model performance for production use	Sprint 2 CNN
		Scott	Evaluate the backup mathematical model's viability against the primary AI predictions	Sprint 2 Models
Sprint 4: Test & Deploy	Validation & Release	Earl	User Acceptance Testing (UAT) for Dashboard features	Validated Code
		Isla	Final Deployment of Web App and Database to production environment	Integrated System

		Nikhil	Validate model accuracy against historical fire data.	Sprint 3 Models
		Scott	Documentation of the neural network approach vs. the backup mathematical methods.	